

Auto Ticket Classification using Flow Designer

A no-code ServiceNow automation for a school IT helpdesk

Problem Statement

Every day the school IT helpdesk receives many incident requests from students and teachers - Wi-Fi outages, projector faults, password resets, sluggish computers and similar problems. At present, IT staff read each request by hand and decide its category, a routine that eats up time and does not keep pace as request volumes grow.

This project removes that manual step. Using ServiceNow Flow Designer, the solution scans the keywords in an incident's short description and description fields and classifies the ticket automatically.

Objectives

- Classify incidents automatically the moment they are created.
- Cut down the manual workload of IT agents.
- Make ticket routing faster and more consistent.
- Deliver a no-code solution that is simple to maintain and extend.

PHASE 1: Requirement Analysis & Planning

Students and teachers regularly raise support requests about network connectivity, hardware failures, account access and system performance. Under the current process, IT staff must open every ticket and:

- Work out what type of issue it is.
- Choose the correct category and subcategory.
- Let the caller know that a ticket has been logged.

This hands-on approach is slow, prone to mistakes and hard to scale as the number of requests rises.

Activity 1: Business Use Case

Business Requirements

The system must be able to:

- 1 Classify IT tickets automatically from the issue description.
- 2 Fill in both Category and Subcategory without human involvement.
- 3 Support dependent-choice behaviour between Category and Subcategory.
- 4 Email the caller automatically when a ticket is created.
- 5 Keep ticket data in a structured, standardised format.
- 6 Remain easy to maintain and ready for future growth.

Activity 2: Creating a New Update Set

Navigation steps:

- 1 Open the **All** menu and search for **Update Set**.
- 2 Under **System Update Sets**, choose **Local Update Sets**.
- 3 Click the **New** button at the top right of the page.
- 4 Enter the name **Project Update Set**.
- 5 Set State to **In progress**.
- 6 Set Application to **Global**.
- 7 **Save** the form.
- 8 In Related Links, click **Make This My Current Set**.
- 9 The new update set now shows as the current one.

Activity 3: Navigation Steps - Picture View

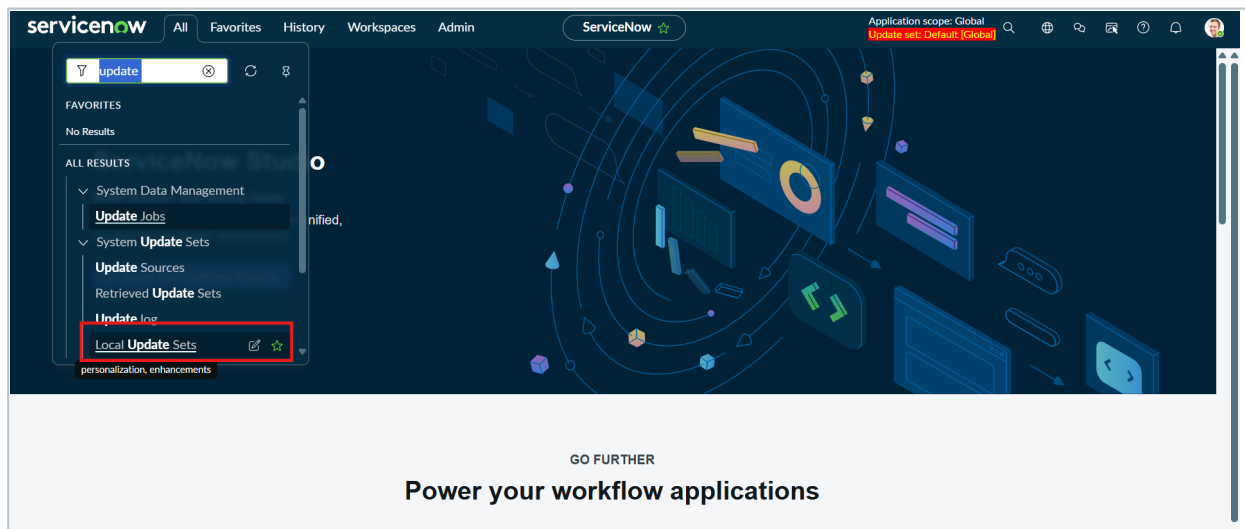


Figure 1: Searching for "update" in the application navigator

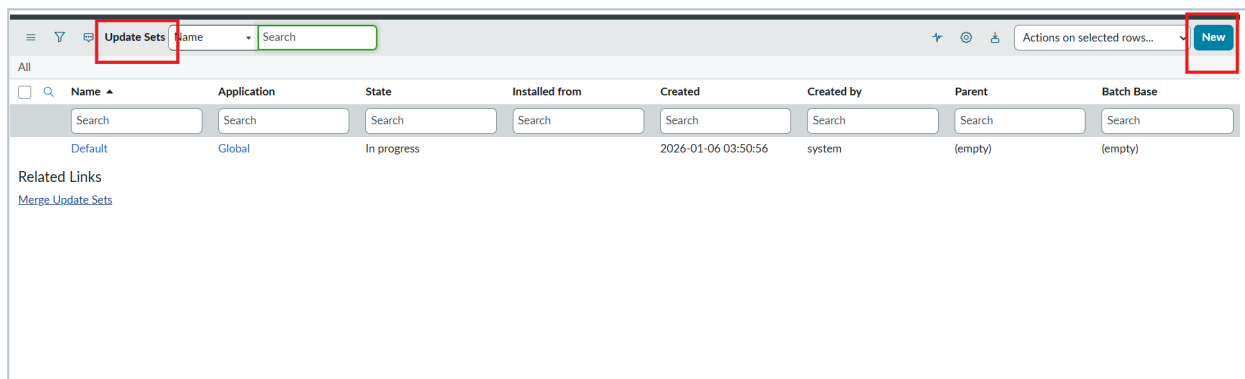


Figure 2: Local Update Sets list - click New to create one

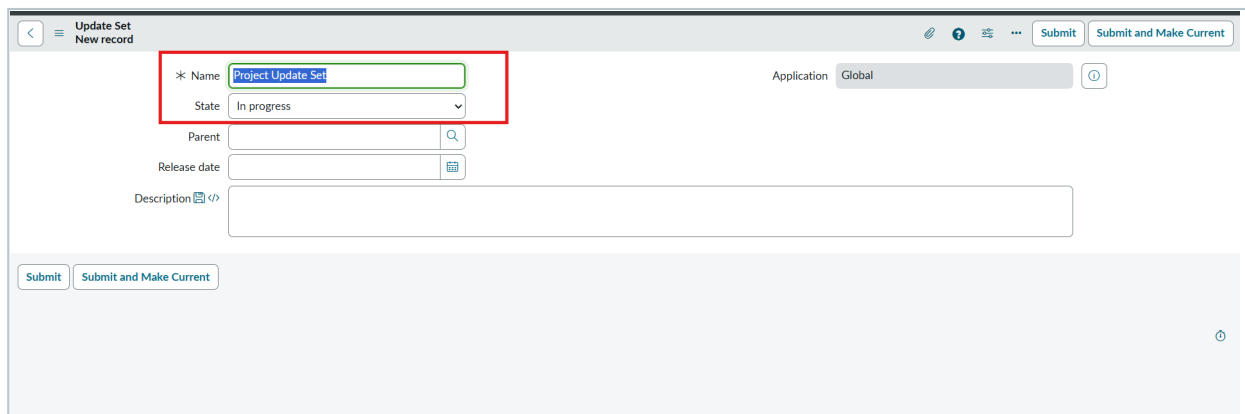


Figure 3: New update set form with name, state and application

Update Set
Project Update Set

* Name: Application: Global

State: Created: 2026-02-06 20:55:31

Parent: Created by: admin

Release date: Merged to:

Install date:

Installed from:

Description

Related Links

[Make This My Current Set](#)

[Merge With Existing Update Set](#)

[Scan Update Set](#)

[\[SN Utils\] Versions / IO](#)

Customer Updates | Update Set Logs | Child Update Sets | Install History

Created Search

Update set = Project Update Set

Created	Type	View	Target name	Updated by	Remote update set	Action
---------	------	------	-------------	------------	-------------------	--------

Figure 4: Saved update set with "Make This My Current Set" in Related Links

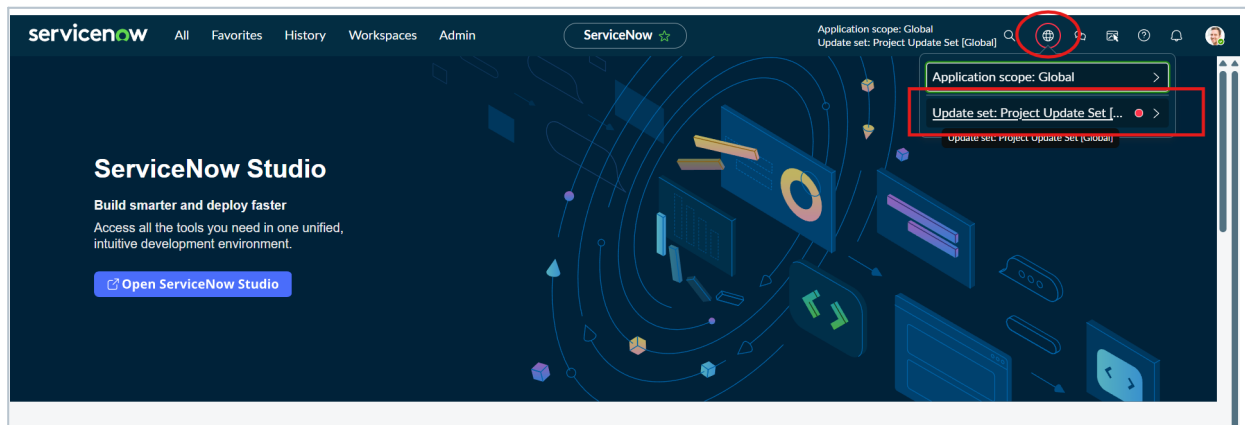


Figure 5: Project Update Set confirmed as the current update set

PHASE 2: Backend Development & Configuration

Activity 1: Creating a Custom Table for Ticket Records

Steps to create the custom table:

- 1 Open the **All** menu and search for **Tables**.
- 2 Under **System Definition**, click **Tables**.
- 3 Click the **New** button at the top right.
- 4 Enter the label **Incident WorkFlow**.
- 5 Untick **Create module**.
- 6 **Save** the form.
- 7 Open the **Controls** tab and tick **Auto-number**.
- 8 **Save** the form once more.

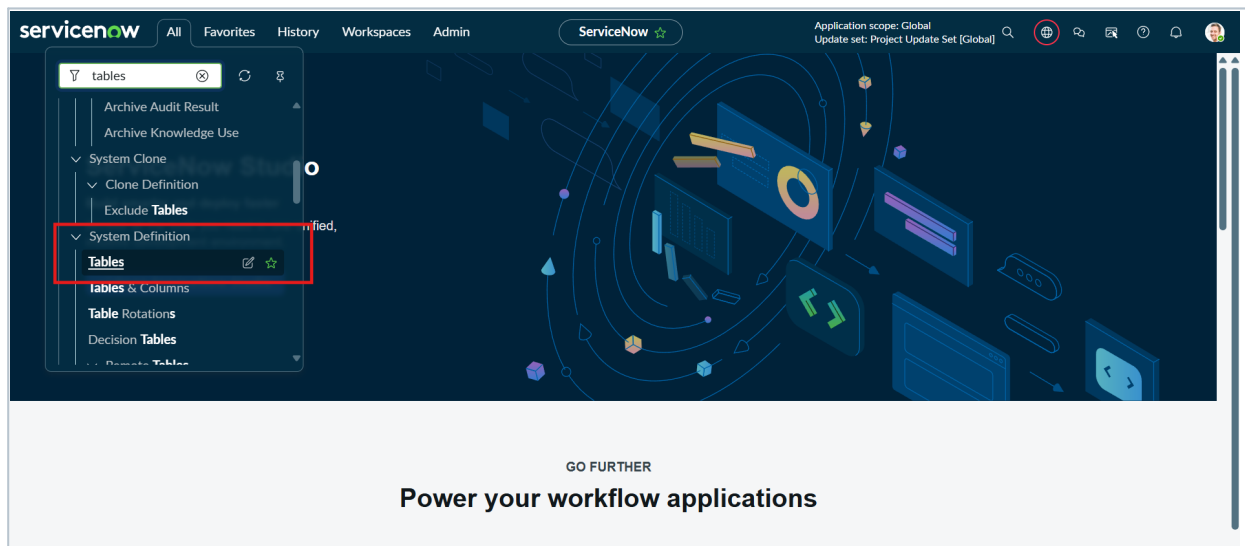


Figure 6: Locating Tables under System Definition

Label	Name	Extends table	Extensible	Updated
Search	Search	Search	Search	Search
Logged in User	v_user_session	(empty)	false	2007-07-13 16:15:23
System Plugin	v_plugin	(empty)	false	2007-10-09 12:10:05
Active Transactions	v_transaction	(empty)	false	2008-05-23 17:30:17
Node Log	v_logfiles	(empty)	false	2009-05-14 17:05:41
Customer Uploads	v_customer_uploads	(empty)	false	2009-05-14 18:38:13
MySQL Process List	v_mysql_proclist	(empty)	false	2012-02-25 19:47:02
MySQL Status	v_mysql_status	(empty)	false	2012-02-25 19:47:08
MySQL Variables	v_mysql_variables	(empty)	false	2012-02-25 19:47:12
IOStats	v_iostats	(empty)	false	2012-02-27 17:13:51
Cluster Nodes	v_cluster_nodes	(empty)	false	2012-02-27 23:48:26
SQL Debug	v_sql_debug	(empty)	false	2012-02-28 16:40:55
Workflow Validation Report	v_wf_validation_report	(empty)	false	2012-10-01 10:43:05
Hyper-V Resource Pool Component	cmdb_ci_hyper_v_rpool_comp	Hyper-V Object	false	2012-10-30 08:07:25
Application Menu	sys_app_application	Application File	false	2012-10-30 13:59:18
Menu Category	sys_app_category	Application File	false	2012-10-30 13:59:44
Scriptable Object	v_scriptable_object	(empty)	false	2012-11-09 15:15:23

Figure 7: Tables list - click New to start a table

ServiceNow recommends creating custom tables in scoped applications. To learn more about creating scoped applications, click [here](#).

A table is a collection of records in the database. Each record corresponds to a row in a table, and each field on a record corresponds to a column on that table. Applications use tables and records to manage data and processes. [More Info](#)

* Label: Incident WorkFlow
 * Name: u_incident_workflow

Application: Global
 Create module: ☐
 Remote Table: ☐

Columns | Controls | Application Access

Table Columns: for text | Search

Column label	Type	Reference	Max length	Default value	Display
Insert a new row...					

Figure 8: New table form with the label and name filled in

Table Incident WorkFlow

Columns Controls Application Access

Extensible ☐

Live feed ☐

Use auto-numbering to define a sequential identifying code made up of a prefix, a base number and a padding value to ensure a consistent format

Auto-number ☒

Prefix INC

Number 500

Number of digits 5

Security Rules (ACLs) are required if anyone other than an administrator needs to work with this table. Creating default security rules will grant full access to this table to anyone with the user role you specify.

Create access controls ☒

* User role u_incident_workflow_user

Delete Update Delete All Records

Related Links

Form Builder

Figure 9: Controls tab with auto-numbering enabled (prefix INC)

- 1 After saving, open **Show Form** from Related Links.
- 2 From the form context menu choose **Configure > Form Design**.
- 3 Add the required fields in the designer.

Incident WorkFlow

Save

Insert

Insert and Stay

Analyze Access

Configure

Export

View

Create Favorite

Copy URL

Copy sys_id

Show XML

History

Reload form

Form Builder

Form Design

Form Layout

Related Lists

All

Table

Security Rules

Business Rules

Client Scripts

UI Policies

Data Policies

UI Actions

Notifications

Dictionary

Number INC00519

Caller Janice Twiet

State -- None --

Assigned group

Assigned to

Update Delete

Figure 10: Opening Form Design from the context menu

Activity 2: Field Creation and Data-Type Configuration

This step creates every field the ticket table needs, each with a suitable data type, choice list and dependency, so that the data captured is structured and consistent.

Field Label	Type	Reference / Choices
Number	Auto Number	-
Caller	Reference	sys_user
Category	Choice	Network, Hardware, Access, Performance
Subcategory	Choice	Wi-Fi, Projector, Forgot Password, Slow Computer

Field Label	Type	Reference / Choices
Short Description	String	-
Description	String	-
State	Choice	New, In progress, On hold, Resolved, Closed
Assigned Group	Reference	sys_user_group
Assigned to	Reference	sys_user

- Create each field with the right type first, then define the choice values in Form Design.
- The choice-list fields are **Category**, **Subcategory** and **State**.
- Add the State choices the same way as the others.
- Click **Save** at the top right, then open the form and reload it.

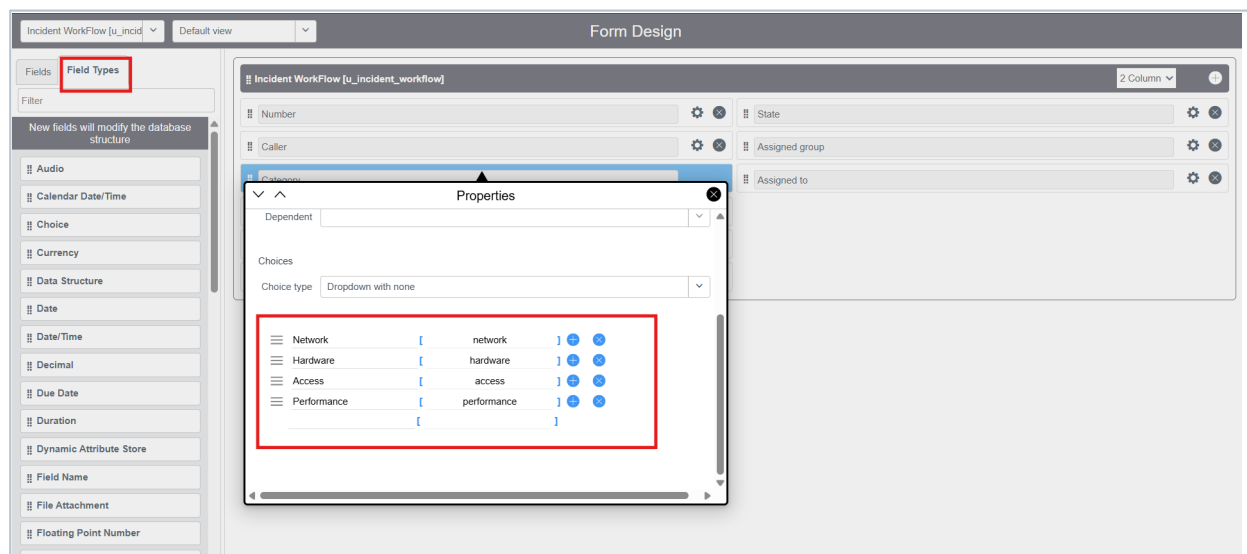


Figure 11: Category choice values defined in Form Design

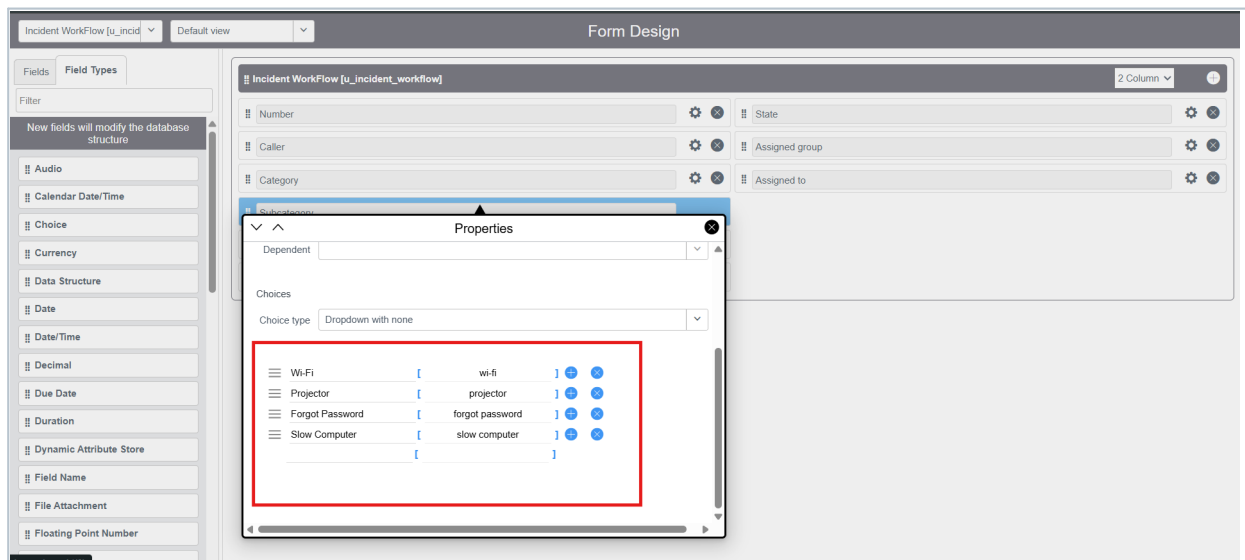


Figure 12: Subcategory choice values defined in Form Design

Activity 3: Dependency Between Category and Subcategory

The goal here is to link the two fields so that the Subcategory options change dynamically depending on the Category that is selected.

Navigation steps:

- 1 Open the form of the new table and right-click the **Subcategory** field.
- 2 Select **Configure Dictionary**.
- 3 In Related Links, open **Advanced view**.
- 4 Go to the **Dependent Field** tab.
- 5 Tick **Use dependent field**.
- 6 Choose **Category** in **Dependent on field**.
- 7 **Save** the record.

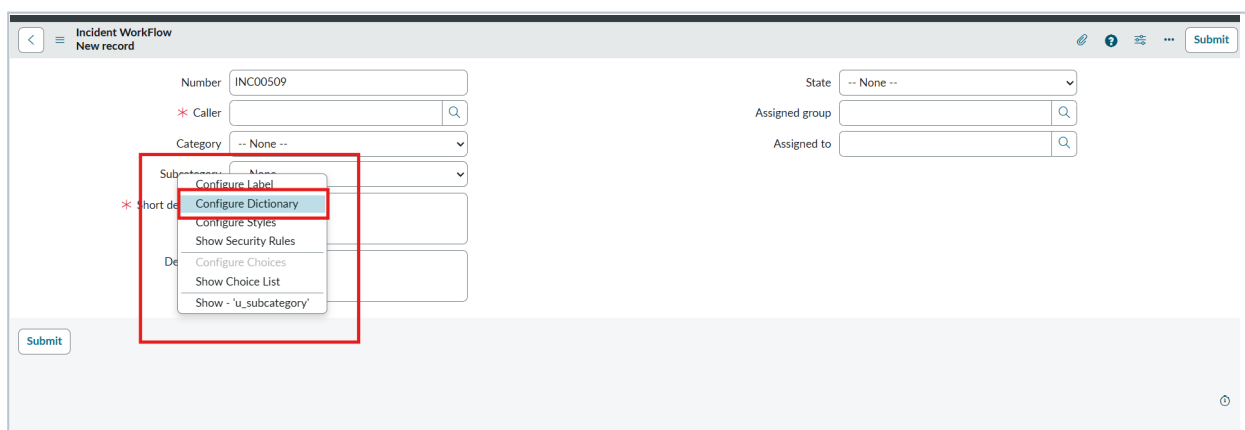


Figure 13: Right-click menu on Subcategory - Configure Dictionary

Dictionary Entry
Subcategory View: Advanced

A dictionary entry manages how ServiceNow stores data in tables and fields (columns). For new dictionary entries, select a **Table** and the field **Type** of the new column. Also enter a column label, which becomes the field label, and the column name. If necessary, set a **Max length** for text String type fields, make the field **Mandatory** to save a record, and make the field a **Display Value** for reference fields so it appears on records that reference this table. [More Info](#)

* Table: Incident Workflow [u_incident_workflow]
 * Type: Choice
 * Column label: Subcategory
 * Column name: u_subcategory

Application: Global
 Active: ☒
 Function field: ☐
 Read only: ☐
 Mandatory: ☐
 Display: ☐

Alters the behavior of a field or functionality that depends on the field. [More Info](#)

Attributes

Choice List Specification: **Dependent Field** | Calculated Value | Default Value

Specify a field whose value is used to determine choice list options for this field. [More Info](#)

Use dependent field: ☒
 Dependent on field: Category

Figure 14: Dictionary entry: dependent field set to Category

Next, scroll to the Choices section and set each Subcategory choice's dependent value to the back-end value of its parent Category:

Subcategory	Dependent on Category
Wi-Fi	Network
Projector	Hardware
Forgot Password	Access
Slow Computer	Performance

Save the record. From now on, the form shows only the subcategory that matches the chosen category.

Dictionary Entry Subcategory View: Advanced

Specify a field whose value is used to determine choice list options for this field. [More Info](#)

Use dependent field ☒

Dependent on field Category

Create Choice List Delete Column Update

Related Links

[Show Table](#)

[Run Point Scan](#)

[\(SN Utils\) Versions \(0\)](#)

Access Controls Choices (4) Attributes Labels (1)

Label Search

Actions on selected rows... New

Label	Value	Language	Sequence	Inactive	Updated	Dependent value
Forgot Password	forgot password	en	2	false	2026-02-06 21:45:02	access
Wi-Fi	wi-fi	en	0	false	2026-02-06 21:44:50	network
Slow Computer	slow computer	en	3	false	2026-02-06 21:45:15	performance
Projector	projector	en	1	false	2026-02-06 21:45:36	hardware

Insert a new row...

Figure 15: Dependent values assigned to each subcategory choice

Incident WorkFlow New record

Number INC00509

State -- None --

* Caller

Assigned group

Assigned to

Category Network

Subcategory -- None --

* Short description -- None --

Wi-Fi

Description

Submit

Figure 16: Category = Network offers only Wi-Fi

Incident WorkFlow New record

Number INC00509

State -- None --

* Caller

Assigned group

Assigned to

Category Hardware

Subcategory -- None --

* Short description -- None --

Projector

Description

Submit

Figure 17: Category = Hardware offers only Projector

Incident WorkFlow
New record

Number: INC00509

State: -- None --

* Caller:

Assigned group:

Category: Access

Assigned to:

Subcategory: -- None --

* Short description:

Description:

Submit

Figure 18: Category = Access offers only Forgot Password

Incident WorkFlow
New record

Number: INC00509

State: -- None --

* Caller:

Assigned group:

Category: Performance

Assigned to:

Subcategory: -- None --

* Short description:

Description:

Submit

Figure 19: Category = Performance offers only Slow Computer

PHASE 3: Automation using Flow Designer & Email Notification

When a student raises a new IT ticket, they only need to provide the **Caller** and a **Short Description** of the problem. There is no need to pick a Category or Subcategory. As soon as the ticket is submitted, a Flow Designer flow starts and checks the short description for predefined keywords such as WiFi, Projector, Password or Slow, then fills in the matching Category and Subcategory.

Classification rules:

If the text contains...	Category	Subcategory
WiFi (or Network)	Network	Wi-Fi
Projector	Hardware	Projector
Password or Login	Access	Forgot Password
Slow or Hanging	Performance	Slow Computer

This automation gives consistent classification, removes human error and greatly reduces the manual effort of IT staff. Because everything is built in Flow Designer, the solution is easy to maintain, scalable and completely no-code.

Activity 1: Creating the Auto-Classification Flow

- 1 Open the **All** menu and search for **Flow Designer**.
- 2 Under **Process Automation**, open Flow Designer.
- 3 At the top right, click **New** and choose **Flow**.
- 4 Flow name: **Auto Classify School IT Tickets**.
- 5 Application: **Global**.
- 6 Click **Build flow**.

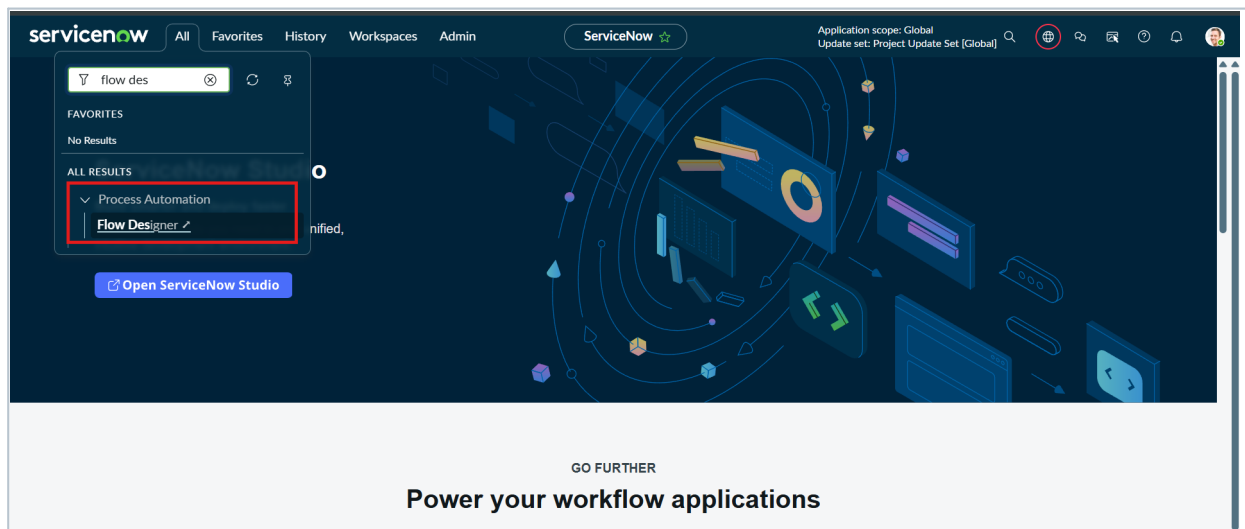


Figure 20: Opening Flow Designer from the navigator

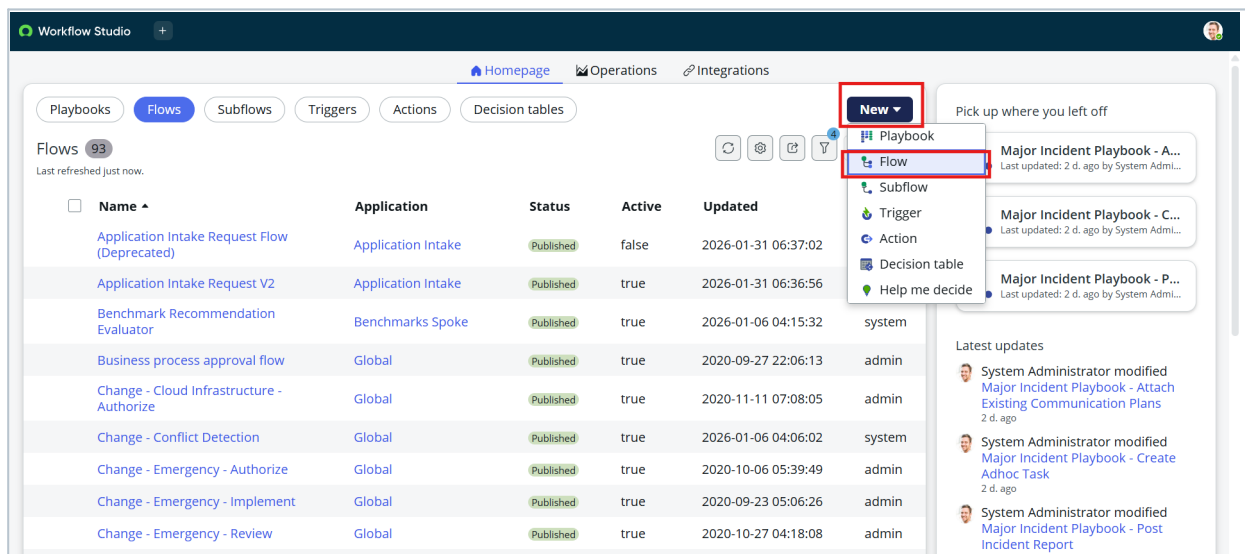


Figure 21: New > Flow in the Workflow Studio home page

Let's get the details for your flow

Flow name *

Application *

Description

[> Show additional properties](#)

Figure 22: Flow details: name and application

Activity 2: Trigger Configuration

- Trigger: **Created** (record created).
- Table: **Incident WorkFlow**.
- Condition: **Category is empty**.
- Click **Done**, then add a step from **Actions > Flow Logic**.

Workflow Studio Auto Classify School IT Tickets

Auto Classify School IT Tickets Inactive

TRIGGER

Incident WorkFlow Created where (Category is empty)

Trigger Created

* Table Incident WorkFlow [u_incident_w...]

Condition All of these conditions must be met

Category is empty

New Criteria

Advanced Options

Delete Cancel Done

Data Collapse All

Flow Variables

Trigger - Record Created

Incident WorkFlow Record Record

Incident WorkFlow Table Table

Run Start Time UTC Date/Time

Run Start Date/Time Date/Time

Figure 23: Trigger set to Record Created with Category is empty

Activity 3: Wi-Fi Issue Classification

- Add an **If** condition and label it, for example, "SD is Wifi OR Network" (any label works).
- Condition 1: with the data pill picker choose **Short description**, operator **contains**, value **Wi-Fi**.
- Condition 2: **Short description contains Network**.
- Click **Done**.
- Add the action **Update Record**.

- Drag the trigger record from the data panel into the Record field; the table fills in as Incident Workflow.
- Set **Category = Network** and **Subcategory = Wi-Fi**, then click **Done**.

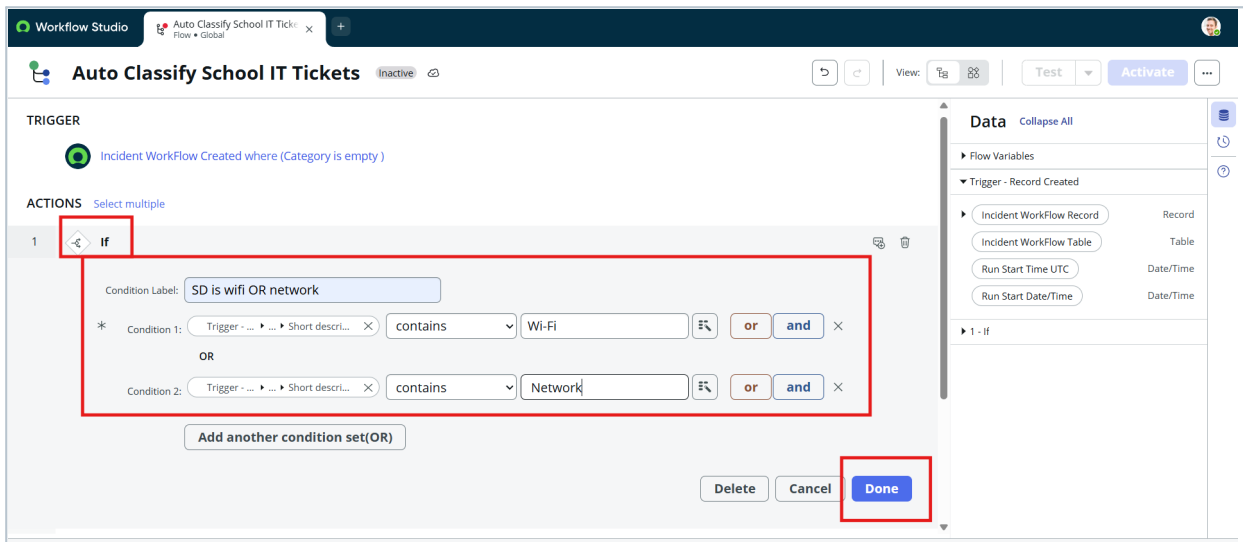


Figure 24: If condition checking for Wi-Fi or Network

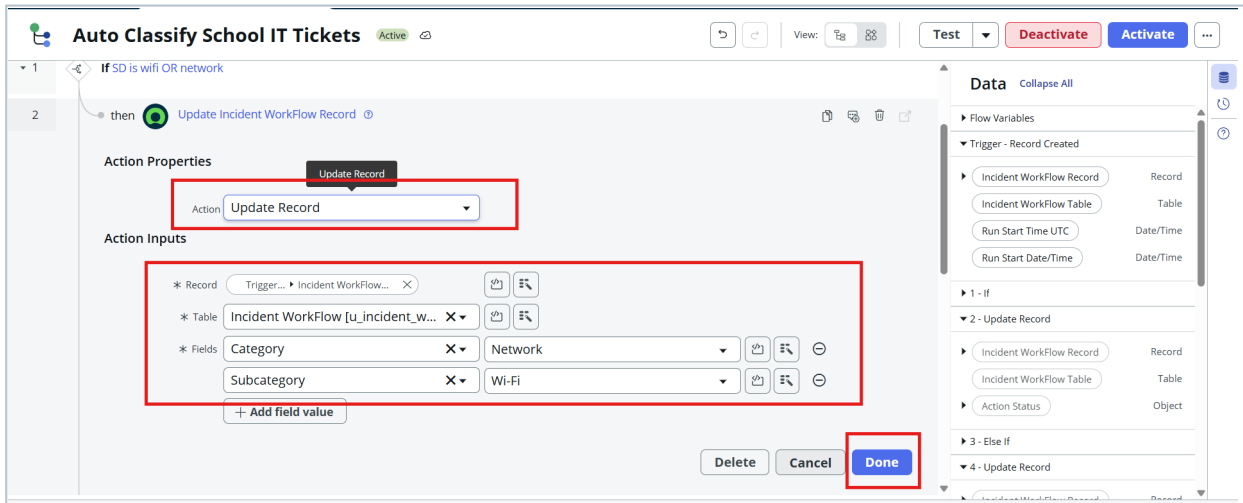


Figure 25: Update Record action setting Network / Wi-Fi

Activity 4: Projector Issue Classification

- Add an **Else If** block labelled "SD is projector OR hardware".
- Condition: **Short description contains Projector**; click **Done**.
- Action: **Update Record** with **Category = Hardware** and **Subcategory = Projector**; click **Done**.

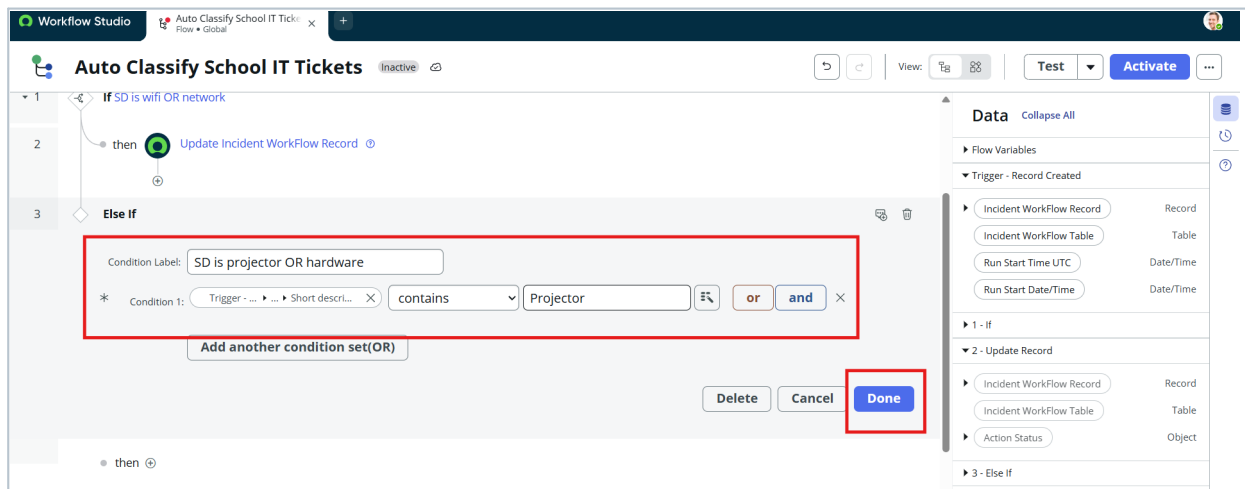


Figure 26: Else If condition for Projector

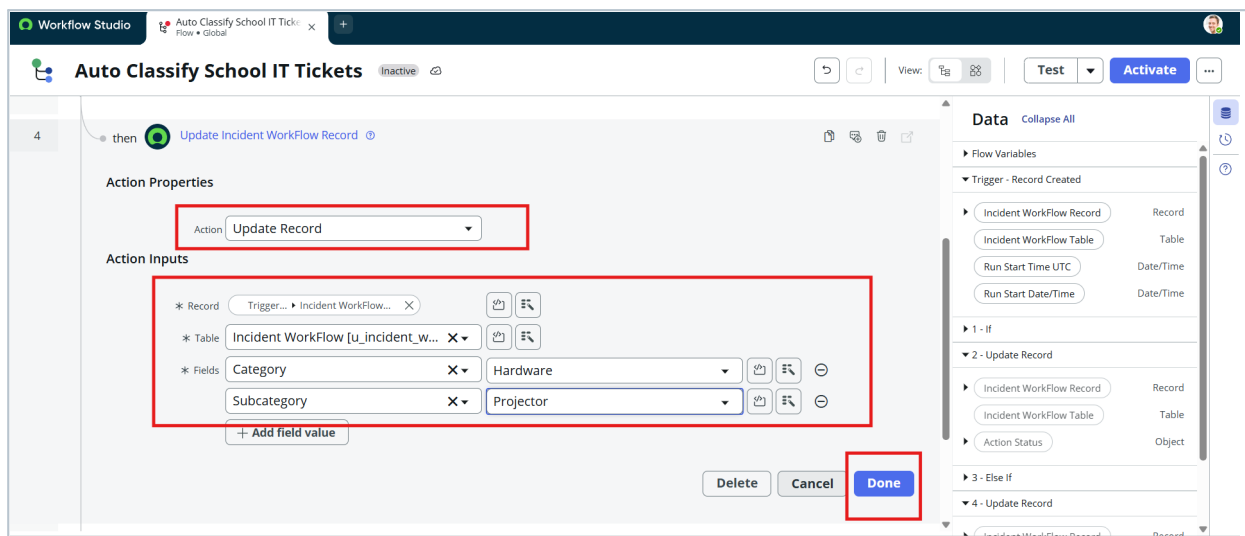


Figure 27: Update Record action setting Hardware / Projector

Activity 5: Password Issue Classification

- Add another **Else If** labelled "SD is Forgot password".
- Condition: **Short description contains Forgot password**; click **Done**.
- Action: **Update Record** with **Category = Access** and **Subcategory = Forgot Password**; click **Done**.

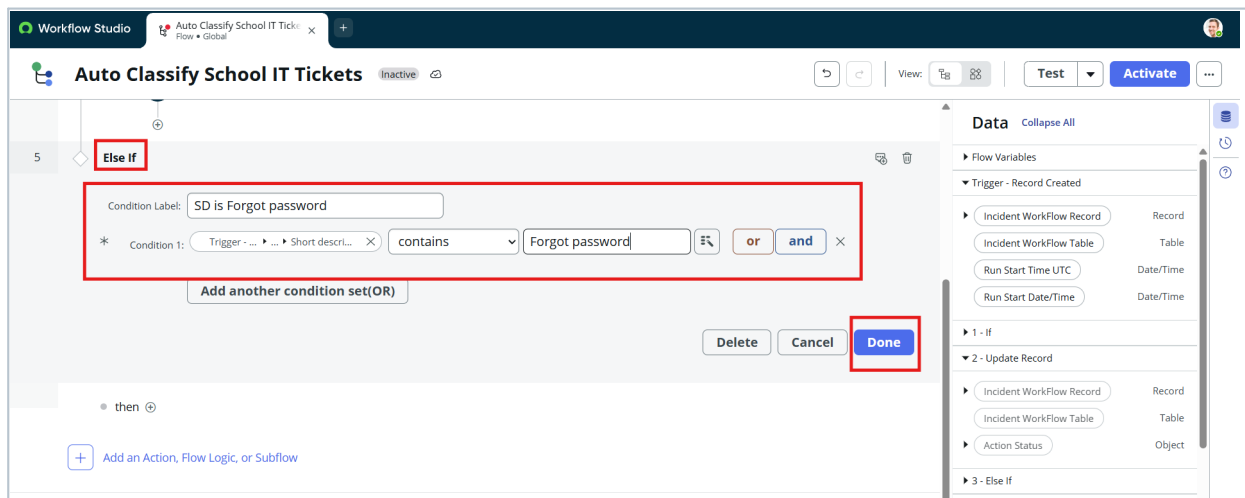


Figure 28: Else If condition for Forgot password

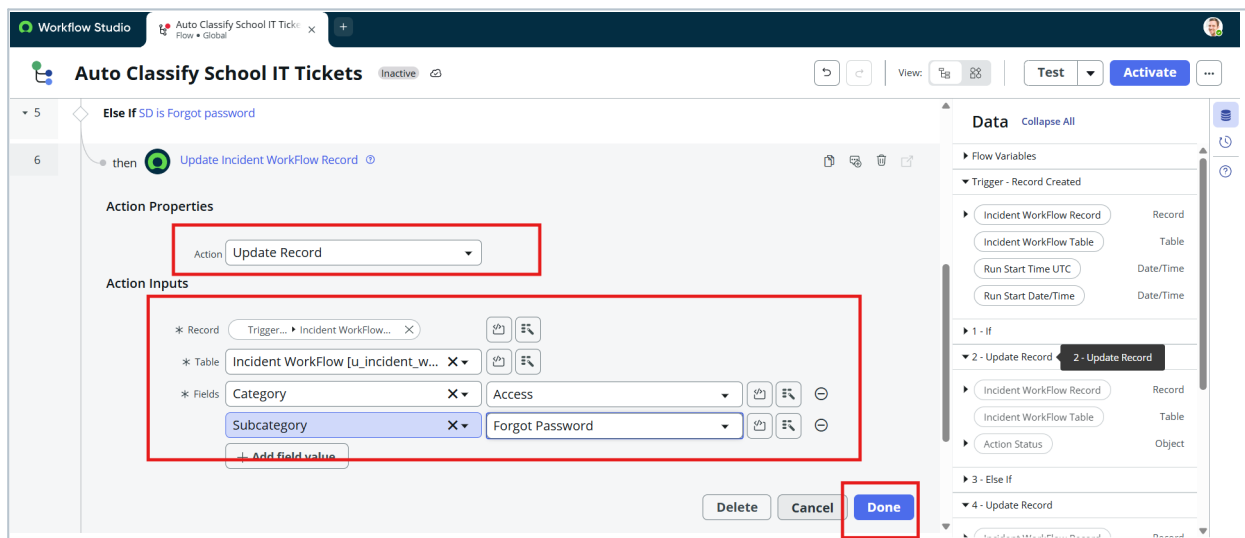


Figure 29: Update Record action setting Access / Forgot Password

Activity 6: Slow Computer Classification

- Add one more **Else If** labelled "SD is Performance".
- Condition: **Short description contains Slow computer**; click **Done**.
- Action: **Update Record** with **Category = Performance** and **Subcategory = Slow Computer**; click **Done**.

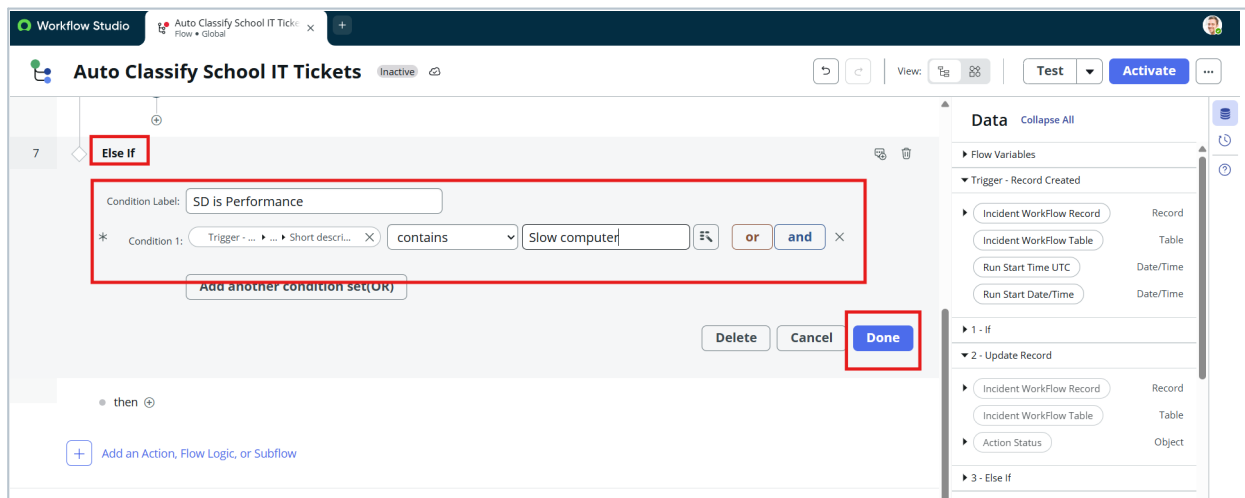


Figure 30: Else If condition for Slow computer

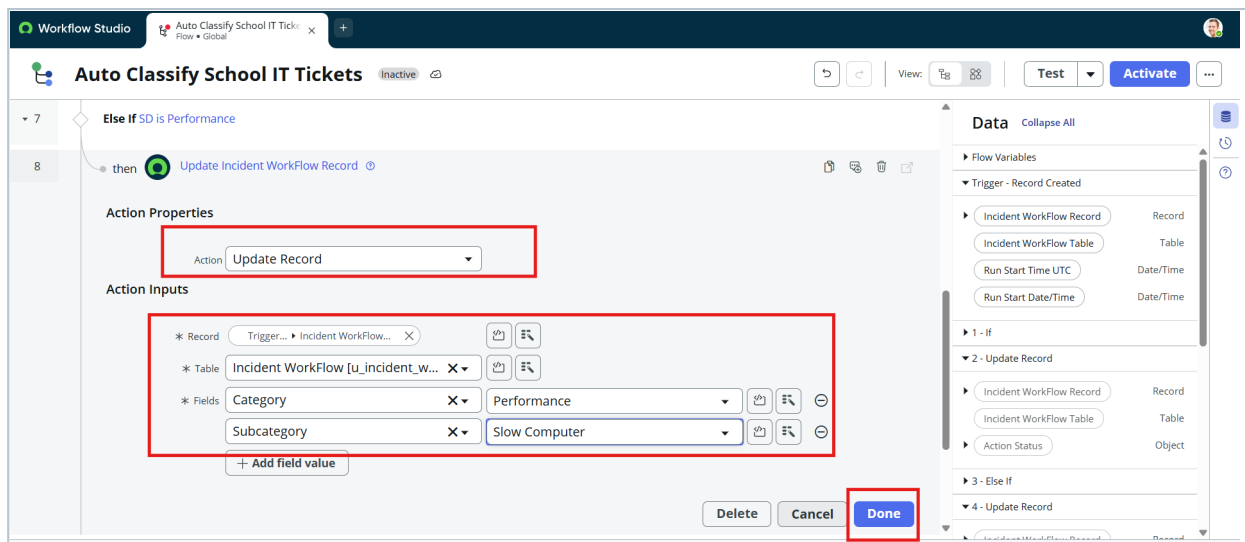


Figure 31: Update Record action setting Performance / Slow Computer

Activity 7: Email Action Setup

Once the ticket is classified, the flow sends a confirmation email to the caller.

- Add the action **Send Email**.
- To: **Caller > Email** (from the data pill picker).
- Subject: **Your Request for the issue has been submitted**.
- Body: a ticket confirmation message.
- Click **Done**.

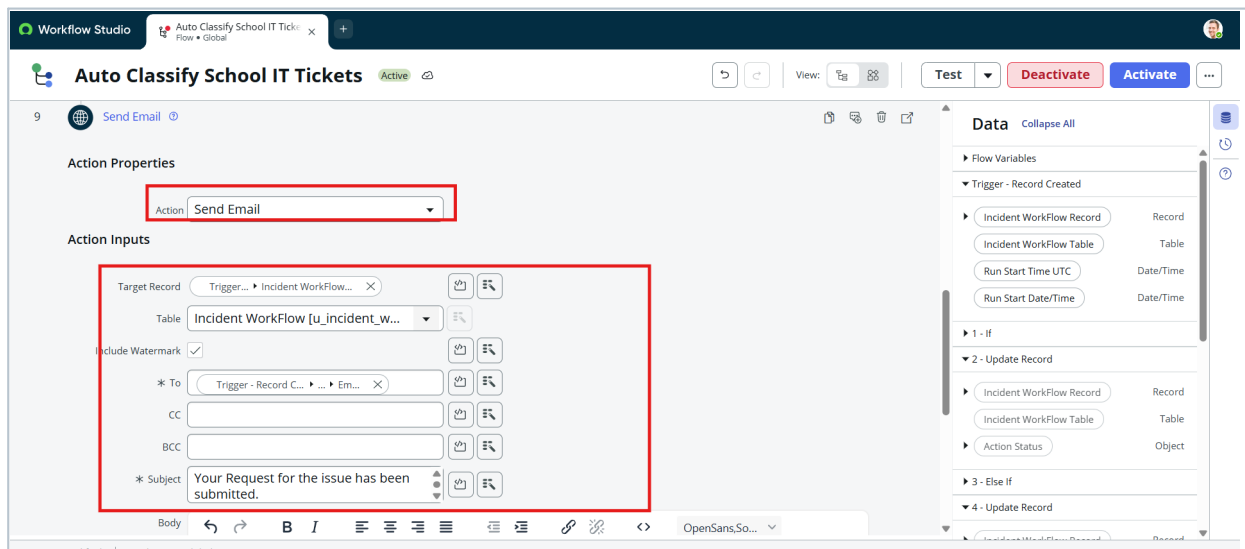


Figure 32: Send Email action configuration

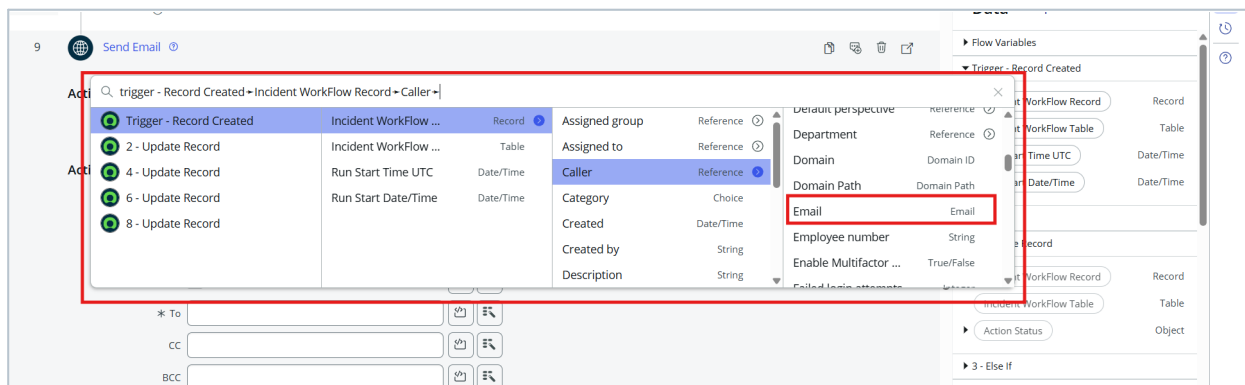


Figure 33: Selecting Caller > Email through the data pill picker

Finally, click **Activate** at the top right to turn the flow on.

PHASE 4: Testing, Validation & Security

This phase confirms that the solution behaves as expected, meets every functional requirement and follows basic security and access-control practice. It validates the automation end to end, checks the accuracy of the data, and ensures that only authorised users can view or change ticket records.

Data Validation Checks

- Mandatory fields are captured correctly.
- Auto-numbers are generated with no duplicates.
- Category and Subcategory values are stored accurately.
- Reference fields resolve correctly against the user and group tables.

Activity 1: Testing & Validation

Test Scenario 1

- 1 Choose a Caller.
- 2 Enter the Short Description: *WiFi not working in library.*
- 3 Save the form, then reload it.

Expected result: Category is filled as **Network**, Subcategory as **Wi-Fi**, and an email goes to the caller's address.

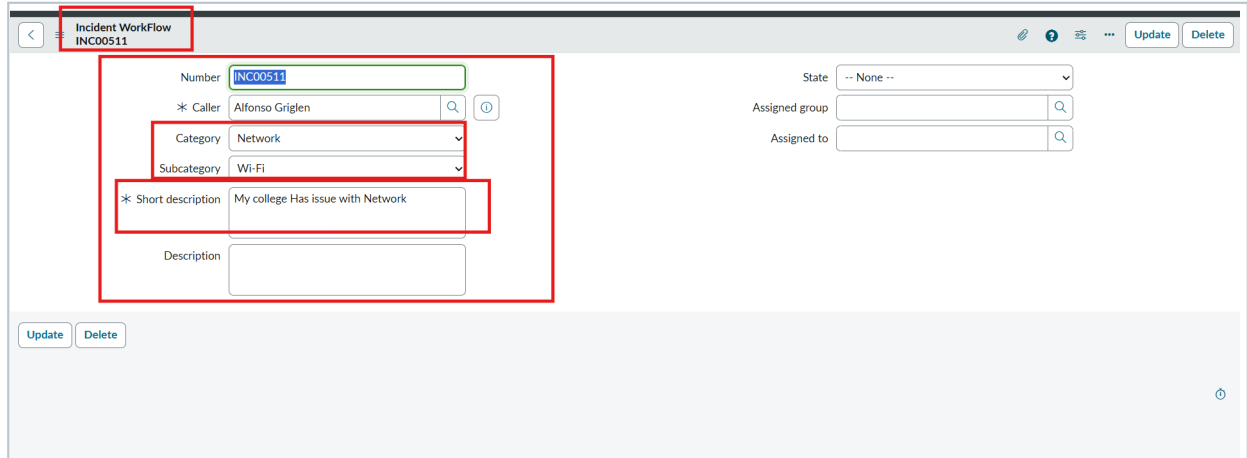
A screenshot of a web application interface for an 'Incident Workflow' form. The form is titled 'INC00511'. It contains several input fields: 'Number' (INC00511), '* Caller' (Alfonso Griglen), 'Category' (Network), 'Subcategory' (Wi-Fi), '* Short description' (My college Has issue with Network), and 'Description'. The 'Category' and 'Subcategory' fields are highlighted with red boxes, indicating they were automatically populated. To the right of the form, there are fields for 'State' (set to '-- None --'), 'Assigned group', and 'Assigned to'. At the bottom left of the form are 'Update' and 'Delete' buttons. At the bottom right of the page is a small circular icon.

Figure 34: Test 1 - ticket automatically classified as Network / Wi-Fi

Activity 2: Checking the Email Notification

- 1 Open the **All** menu and search for **Emails**.
- 2 Under **System Logs**, select **Emails**.
- 3 Search using the subject line.
- 4 Open the record and click **Preview Email** in Related Links.

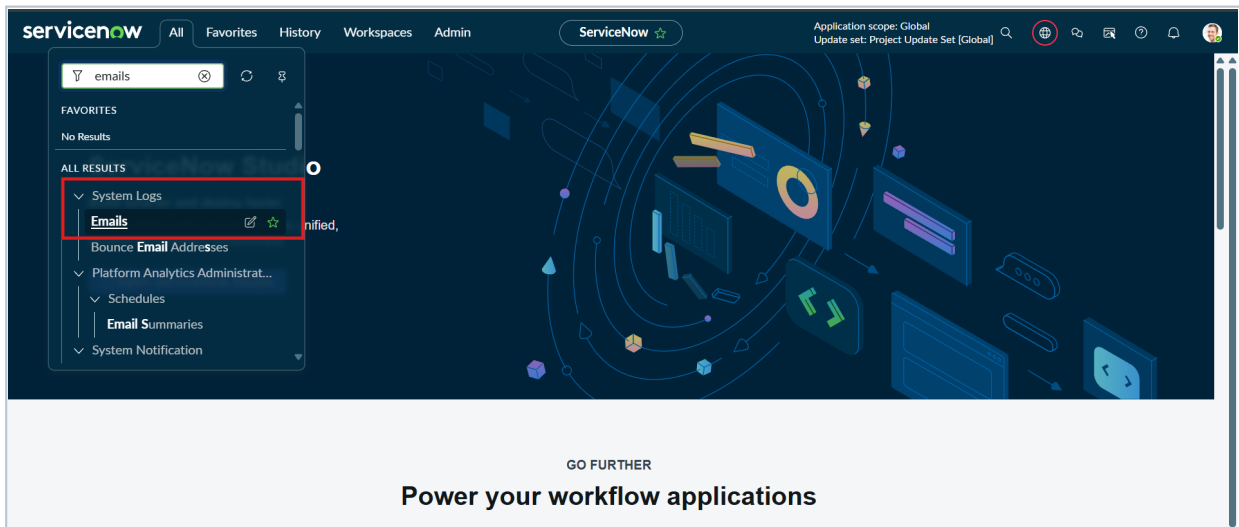


Figure 35: Locating Emails under System Logs

The screenshot shows the 'Emails' list view in ServiceNow. The table has columns for 'Created', 'Recipients', 'Subject', 'Type', 'Notification type', and 'User ID'. A red box highlights the 'Subject' column. The data rows show two email entries with the subject 'Your Request for the issue has been submitted.'.

Created	Recipients	Subject	Type	Notification type	User ID
2026-02-06 23:04:59	savannah.kesich@example.com	Your Request for the issue has been submitted.	send-ready	SMTP	(empty)
2026-02-06 23:04:52	eliseo.wice@example.com	Your Request for the issue has been submitted.	send-ready	SMTP	(empty)

Figure 36: Email log filtered by the confirmation subject

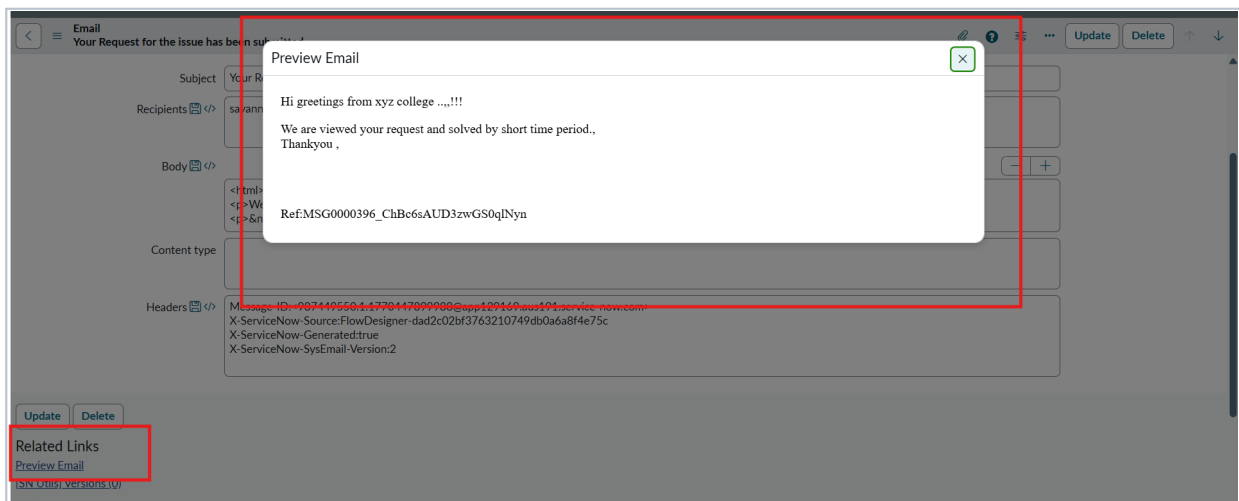


Figure 37: Preview of the email received by the caller

Activity 3: Test Scenario 2

- 1 Choose a Caller.
- 2 Enter the Short Description: *Projector not turning on.*
- 3 Save the form, then reload it.

Expected result: Category is filled as **Hardware**, Subcategory as **Projector**, and the caller receives an email.

Incident WorkFlow
INC00512

Number: INC00512

* Caller: Sam Sorokin

Category: Hardware

Subcategory: Projector

* Short description: Projector not turning on

Description:

State: -- None --

Assigned group:

Assigned to:

Update Delete

Figure 38: Test 2 - ticket automatically classified as Hardware / Projector

PHASE 5: Deployment & Final Documentation

This phase moves the finished solution to the target environment and completes the project documentation. It ensures the solution is stable, reusable and ready for demonstration or production use, backed by clear documentation for future reference.

Activity 1: Completing and Exporting the Update Set

- 1 Open the **All** menu and search for **Local Update Sets**.
- 2 Under **System Update Sets**, select **Local Update Sets**.
- 3 Open **Project Update Set**.
- 4 Change State from **In progress** to **Complete**.
- 5 **Save** the form.
- 6 In Related Links, click **Export to XML**.
- 7 The update set is downloaded to your computer as an XML file.
- 8 Share the file by email or WhatsApp - it can then be loaded on another instance.

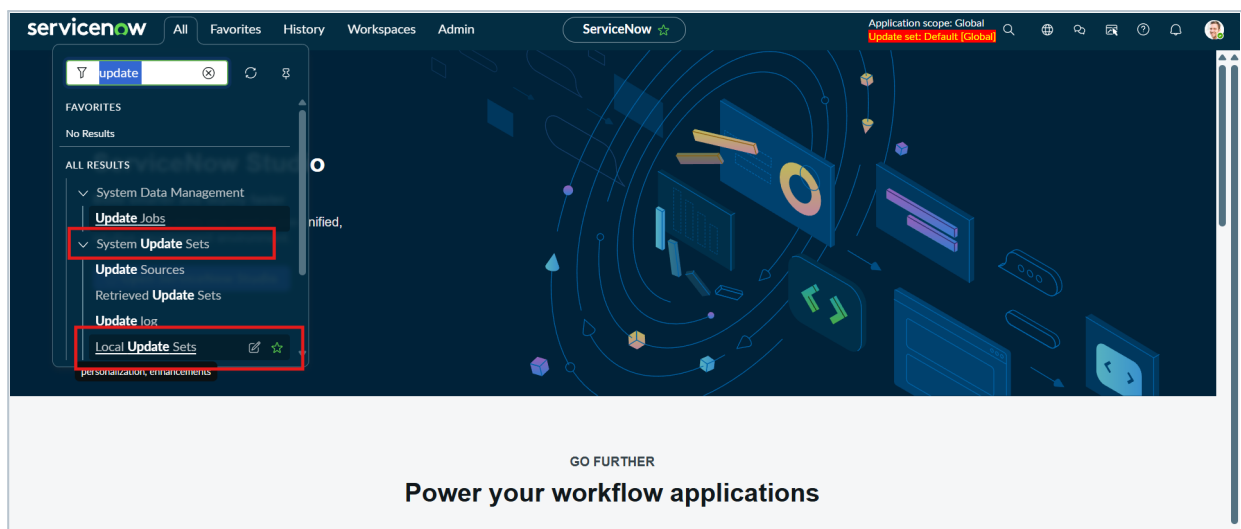


Figure 39: Navigating to Local Update Sets

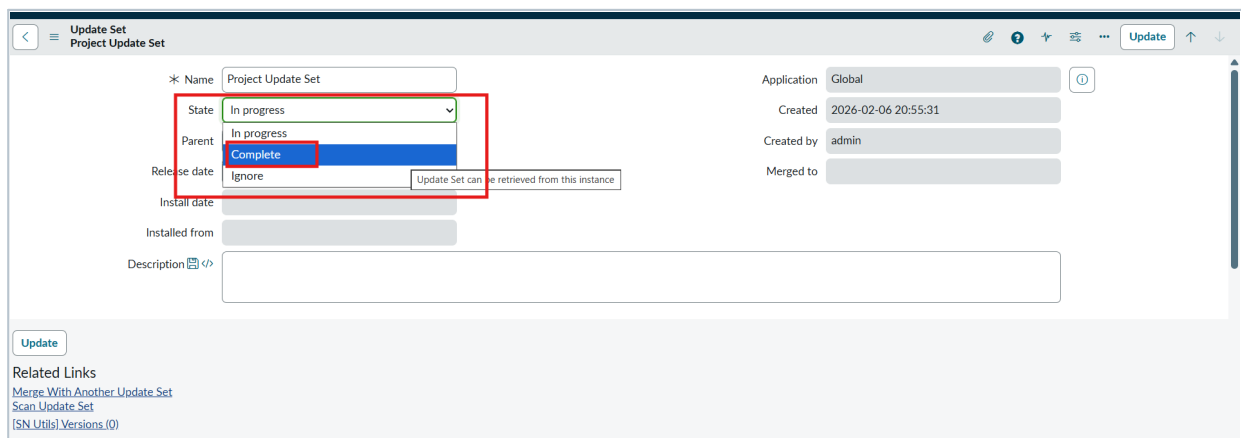


Figure 40: Changing the update set state to Complete

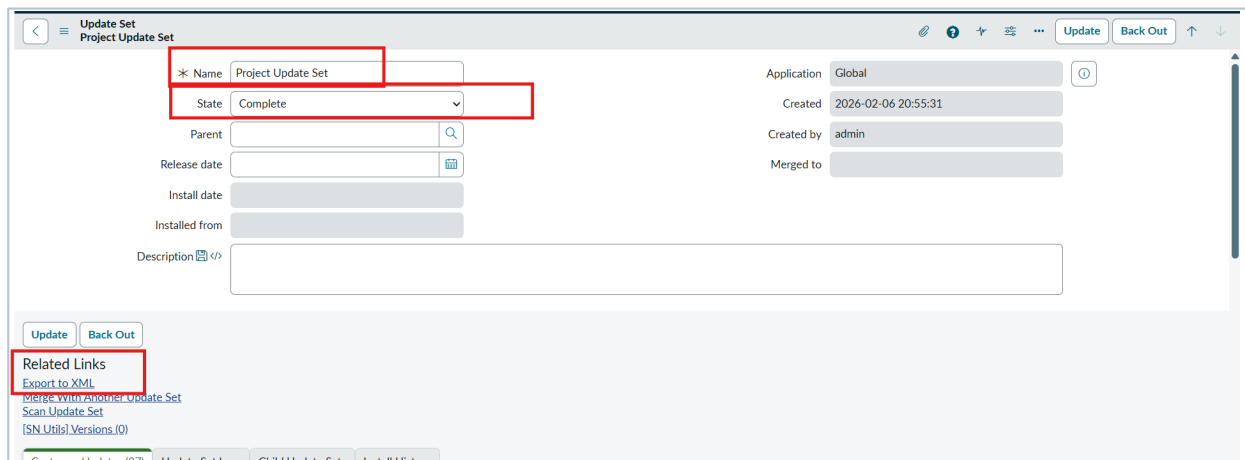


Figure 41: Completed update set with the Export to XML link

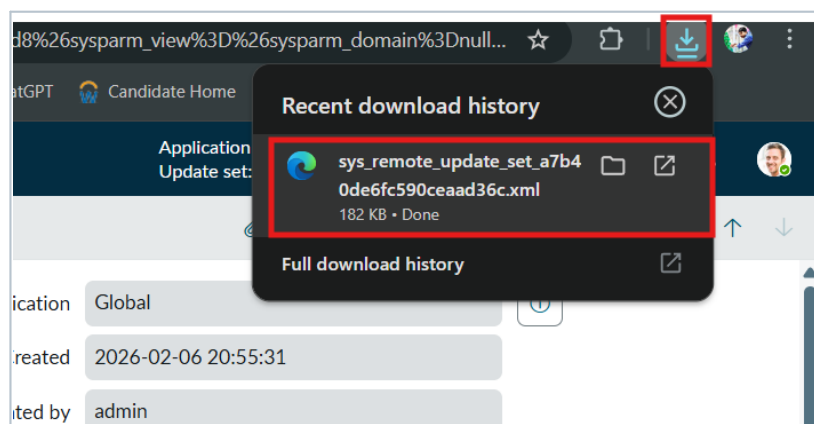


Figure 42: Exported update set XML file in the download history

Conclusion

- The Auto Ticket Classification project delivers a complete, end-to-end automation for a school IT helpdesk. Its main goal - reducing the manual effort of categorising tickets and speeding up responses - has been fully met.
- Through structured requirement analysis, careful backend setup and no-code automation in Flow Designer, tickets are now classified automatically from the short description a student enters. Dependent choice fields keep the data accurate, and automatic email notifications acknowledge each ticket to the caller immediately.
- A phased approach kept the work clear, stable and easy to maintain. Testing and security validation confirmed that the system runs reliably, preserves data integrity and enforces suitable access controls.
- Overall, the project shows how ServiceNow can deliver scalable, maintainable and user-friendly automation without complex scripting or machine learning. It suits small and medium environments such as schools and colleges, and can later be extended with features like automatic assignment, SLA tracking or predictive intelligence.